Ashish Srivastava

ashishsri@proton.me · +1-970-459-0596 · Denver, Colorado
linkedin.com/in/ashish-srivastava-19as

Professional Summary

Data Scientist and Engineering Professional with 4+ years of experience in computational engineering research, statistical modeling, and cross-functional analytics. Proven track record of developing machine learning models, optimizing manufacturing processes, and delivering data-driven insights to C-suite executives. Expert in Python, AWS, statistical analysis, and process optimization with strong background in chemical engineering and data science.

Core Competencies

Nuclear and Plasma Physics: Plasma Reactor Physics, Monte Carlo methods, Heat Transfer Modeling, Statistical Process Control for Reactor Systems
Engineering Systems: Process Optimization, Statistical Process Control, Heat Transfer, Fluid Mechanics, Control Systems
Regulatory and Safety: Nuclear Energy Research, Policy Analysis, Government Relations, Public Relations, Technical Communication, OSHA, Environmental Health and Safety
Programming and Simulation: Python, MATLAB, Fortran, VBA, Bash, SQL, AWS (S3, EC2, Lambda), High Performance Computing, Linux
Data Science: Machine Learning, Statistical Analysis, Computer Vision, Time Series Analysis, Regression Modeling
Tools: JMP, MiniTab, SAP, Jupyter Notebooks, Microsoft Office Suite, LaTeX, SharePoint, Git

Professional Experience

Engineering Data Scientist — Summer 2023 – Spring 2025

Monolith (DOE-Funded Plasma Pyrolysis Startup), Denver, Colorado

- Developed statistical models and process control systems for industrial plasma reactors in DOE-funded clean energy startup environment
- Built plasma voltage model using monte carlo simulation techniques and ensemble methods, eliminating need for live reactor monitoring
- Built computer vision model for automated quality control and predictive maintenance, reducing manual inspection time by 75%
- Architected AWS data pipelines using S3, EC2, and Lambda services
- Created internal Python libraries that streamlined analytical workflows and reduced analysis time by 60%
- Collaborated with cross-functional stakeholders including capital markets, R&D, and process engineering teams to deliver data-driven insights using statistical analysis and cloud infrastructure
- Translated complex statistical investigations into actionable insights for C-suite executives and operations teams

STEM Policy Legislative Aide — Summer 2021

Office of Senator Michael Bennet, Washington, DC

- Only STEM major in legislative aide cohort, bringing technical expertise to policy analysis
- Conducted and communicated condensed research on internet bots being used to influence public sentiment
- Completed personal technical research project on nuclear energy regulation and public sentiment psychology
- Applied engineering analytical skills to policy research and legislative communication

University Lecturer — Spring 2022 – Winter 2023

University of Colorado Boulder, Boulder, Colorado

- Created and delivered lectures for Chemical Engineering courses including chemical engineering heat transfer, fluid mechanics, and computational methods
- Developed comprehensive training materials on high performance computing and terminal interfaces for engineering workflows
- Recorded exam review sessions that improved student performance by 20%
- Designed course content for 200+ students across multiple engineering disciplines

Undergraduate Research Assistant — Summer 2021 – Present

University of Colorado Boulder, Boulder, Colorado

- Co-authored publication in Cambridge Journal of Fluid Mechanics on droplet motion in rectangular channels
- Developed Fortran simulations and Python-based computer vision algorithms for microfluidic systems analysis
- Trained 6 researchers on computational methods and statistical analysis techniques
- Presented research findings at NCUR 2023 and multiple academic conferences
- Designed and built experimental apparatus to validate computational predictions

Project Leader – Data Science Capstone — Spring 2023

CU Boulder CHEN 4838 and Somalogic, Boulder, Colorado

- Led 4-person chemical and biological engineering team in flagship data science course
- Applied computer vision and signal processing techniques to optimize DNA microarray slide manufacturing
- Delivered 15% improvement in manufacturing quality through statistical process optimization

Python Course Developer — Summer 2022 – Spring 2023

University of Colorado Boulder, Boulder, Colorado

- Redesigned freshman engineering computing curriculum, transitioning from Excel/VBA/MATLAB to Python
- Developed automated grading systems and programming assignments for multiple engineering departments
- Impacted 500+ students annually through curriculum improvements
- Created scientific software solutions deployed across engineering programs

Research Team Leader — Fall 2020

CU Boulder PHYS 1140 and LASP, Boulder, Colorado

- Led 4-person interdisciplinary team analyzing solar flare X-ray data from GOES-15 satellite
- Investigated energy-frequency relationships using Jupyter notebooks and linear regression analysis
- Presented findings on space weather patterns and solar activity correlations

Education

Bachelor of Science in Chemical and Biological Engineering **Fall 2019 – Fall 2023**
University of Colorado Boulder, Boulder, Colorado
GPA: 3.76 Minors: Computational Biology, Biomedical Engineering
Thesis: Experimental and Computational Analyses of Droplet Motion in Straight, Rectangular Microchannels
Awards: Fall 2023 Academic Engagement Award, Fall 2023 Outstanding Undergraduate of the College of Engineering

Publications

Co-author, Cambridge Journal of Fluid Mechanics – "Droplet Motion in Rectangular Channels: Experimental and Computational Analysis"

Co-Author, The Astrophysical Journal – "Coronal Heating as Determined by the Solar Flare Frequency Distribution Obtained by Aggregating Case Studies"

Languages

English (Native), Spanish (Intermediate), Hindi (Intermediate), Korean (Intermediate), Urdu (Novice)